TEXORA · LECTURE SHEET

Marker Making

The plan that arranges every pattern piece on the width of the fabric before a single ply is cut — where marker efficiency is defined, calculated and fought for, because this plan decides how much fabric becomes garments and how much becomes waste.

TE-2207 · APPAREL MANUFACTURING-1 · TOPIC 4 · LEVEL 2 TERM 2

1. What a marker is

DEFINITION

A marker is the arrangement of all the pattern pieces of one or more garments — in the sizes and quantities planned for a cutting — fitted within the usable width of the fabric over a definite length. It is drawn or printed at full scale (or stored as a CAD file), laid on top of the spread fabric, and the cutter follows its outlines. The length of the marker is the length of the lay that will be spread beneath it (Topic 5).

Think of the marker as a jigsaw played for money. The pieces are fixed by the pattern room (Topic 3) and cannot change shape; the rectangle they must fill is fixed by the fabric width; only their positions and rotations are free. Every square centimetre of the rectangle is paid for as fabric, so the skill of marker making is measured by how little of the rectangle is left uncovered by pieces.

2. Marker efficiency

DEFINITION

Marker efficiency is the percentage of the total marker area that is covered by the pattern pieces. The remaining percentage is marker waste — fabric that will be cut away between and around the pieces.

$$\text{Marker efficiency (\%)} = (\text{Total area of pattern pieces} \div \text{Total area of the marker}) \times 100$$

$$\text{Marker area} = \text{Marker length} \times \text{Usable fabric width}$$

Worked example. A marker is planned for a fabric of usable width 1.45 m. The marker length is 6.5 m, and the areas of all pattern pieces in it, measured by the CAD system, total 7.92 m².

- Marker area = 6.5 × 1.45 = 9.425 m²
- Marker efficiency = (7.92 ÷ 9.425) × 100 = 84.0%
- Marker waste = 100 − 84.0 = 16.0% of the fabric in the lay — about 1.51 m² in every marker length.

Now the commercial point. If better arrangement lifts the same marker to 88% efficiency, the saving per marker length is (0.88 − 0.84) × 9.425 ≈ 0.38 m² of fabric. Multiplied by the hundreds of lays in an order, a few points of efficiency are worth more than almost any other saving available to a factory — and fabric is the largest cost component of most garments, so efficiency goes straight to the price the factory can quote.

A second figure is taken from the same plan: **fabric consumption per garment**. If the marker above contains the pieces of 4 complete garments, the length consumed per garment is 6.5 ÷ 4 = 1.625 m of 1.45 m-wide fabric — the number costing and fabric ordering are built on.

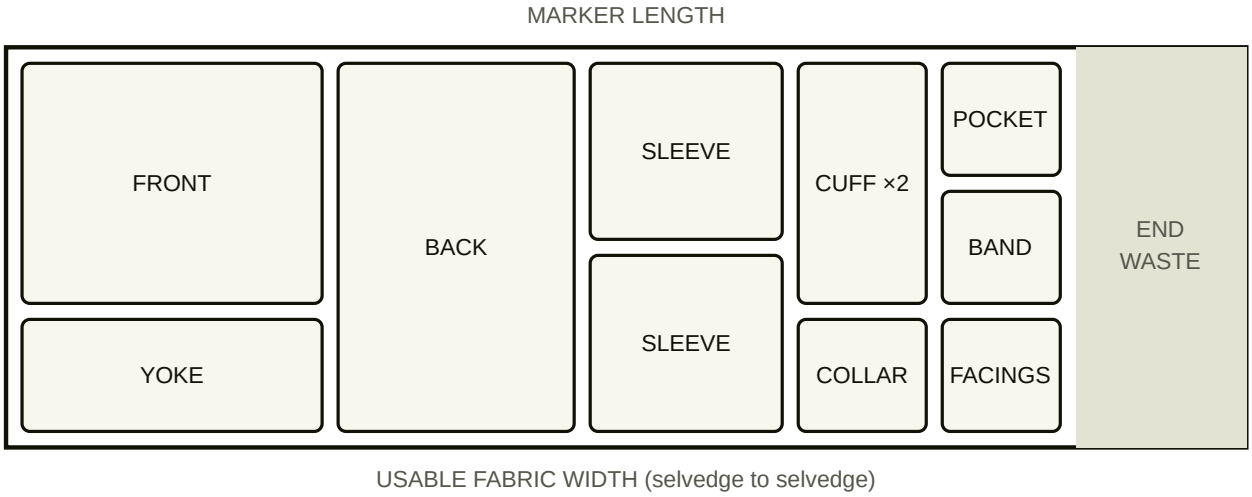


Figure 1 — Schematic marker for a shirt: pieces nested across the fabric width; the uncovered gaps between pieces and at the end are marker waste.

2.1 What efficiency to expect

Well-made markers for woven garments commonly reach about 80–90%. The figure falls when pieces are few and large (a one-size marker of large sizes packs badly), when checks or stripes must be matched, when nap direction forbids turning pieces, and on narrow fabrics. Knit markers often run lower because tubular or relaxed widths vary. An efficiency quoted without its constraints means little — the technologist always asks: for which fabric width, which sizes, and under which matching rules?

3. Objects of marker making

- **To minimise fabric waste** — the primary object, expressed as marker efficiency.
- **To plan the cutting** — the marker fixes the lay length, the number of garments per lay and the size mix in each cutting, so spreading and cutting can be organised from it.

- **To determine fabric consumption** — marker length and efficiency give the consumption per garment used in costing and in fabric order quantities.
- **To guarantee completeness** — a correct marker contains every piece of every garment in the lay, in the right sizes, quantities and mirror pairs; nothing may be discovered missing after the lay is spread.
- **To place pieces correctly on the fabric** — grain lines aligned to the warp, matched checks and stripes positioned, pile direction respected, pieces kept clear of the selvedges and of known fabric faults.

4. Constraints on the marker maker

CONSTRAINT	WHAT IT FORBIDS OR FORCES
Fabric width	No piece may cross the usable width; a piece wider than the fabric forces a seam redesign or a different fabric.
Grain direction	Each piece's grain line must lie parallel to the warp (lengthwise); pieces may rotate 180° but not 90°, unless the pattern allows a cross-grain cut.
Nap / pile direction	On velvet, corduroy and one-way prints, all pieces must point the same way, so the rotation freedom disappears and efficiency drops.
Pattern matching	Checks and stripes must meet at seams; pieces must be placed at positions fixed by the fabric repeat, not wherever they fit.
Complete size sets	The size ratio of the order must be represented as planned; a marker cannot leave out an awkward size to improve its efficiency figure.
Marker length and lay limits	The marker cannot exceed the spreading table length, and longer markers make larger, less flexible lays.
Fabric faults	Known defects are either avoided in placement or the affected pieces are re-cut later from a small repair marker.

5. Methods of marker making

5.1 Manual marker making

The marker maker works at a full-scale table with the paper patterns. The pieces are arranged and re-arranged inside a frame of the fabric width until the arrangement satisfies all constraints at a good efficiency; the outlines are then drawn. Two drawing routes are used:

- **On marker paper** — pieces are traced onto a sheet of marker paper the width of the fabric. The paper marker is laid on the lay and either cut through with the fabric or used as the cutting guide; the drawn outlines must be complete, clean and visible.
- **Directly on the fabric** — pieces are traced with chalk or marker pen onto the top ply of the lay itself. This saves paper but the drawing must wait for each lay, and errors are drawn on the material itself.

5.2 Computerised (CAD) marker making

Digitised patterns are arranged on screen within the fabric width, interactively by an operator or automatically by nesting software, and the finished marker is plotted on paper by a plotter or stored for a CNC cutter (Topic 7). The computer reports piece areas and efficiency exactly, tries far more arrangements than a human can, and stores markers for reuse — which is why CAD has become the standard in export factories.

5.3 Duplication of markers

When several lays must be cut from the same plan, the marker is duplicated rather than re-planned. Established methods: **carbon duplication** (a copy drawn through carbon paper as the original is drawn), the **perforated marker** (the outline is punched as a line of small holes and dusted with powder or chalk through onto the top ply of each lay), and **photographic/diazo copies** of the master marker. Each copy must be checked against the master — a duplicated error repeats in every lay.

6. Wastage in marker making

Marker waste is the fabric inside the marker rectangle that no pattern piece covers. It arises in identifiable places:

- **Inter-pattern waste** — the gaps between curved and angled pieces; the largest share, reduced only by better nesting.
- **End waste** — the unusable remainder where the last pieces do not fill the marker length, and the small end allowance left for lay alignment in spreading.
- **Width (selvedge) loss** — the unusable strips at the edges when pieces cannot use the full width.
- **Matching loss** — the extra fabric consumed because checked/striped pieces are placed by the repeat, not by best fit.

Marker waste must be distinguished from the other fabric losses of the cutting room — spreading end loss, splice loss and defective re-cuts (Topics 5 and 6). A factory that lumps them together cannot see which operation is bleeding fabric. In reporting, marker efficiency measures the plan; the other losses measure the execution.

7. Glossary

TERM	MEANING
Marker	The full-scale plan of pattern pieces within the fabric width over a definite length, guiding the cutting of a lay.
Marker efficiency	Pattern-piece area as a percentage of the total marker area.
Lay	The stack of fabric plies spread to the marker's length, cut together (Topic 5).
Nesting	The fitting of pieces into the marker — the interlocking arrangement itself.
Consumption	The fabric quantity used per garment, derived from marker length and garments per marker.
Block / repeat placement	Placing checked or striped pieces at fixed intervals of the fabric repeat so patterns match at seams.

8. Key points and summary

- A marker is a cutting plan: all pieces of the planned garments fitted inside the usable fabric width over a set length.
- Marker efficiency = pattern area ÷ marker area × 100; worked example: 7.92 m² in a 6.5 × 1.45 m marker = 84.0%.
- Its objects: minimise waste, plan cutting, fix consumption, guarantee complete sets, and place pieces correctly for grain, nap and matching.
- Constraints — width, grain, nap, matching, complete size sets, lay length, fabric faults — define the game; efficiency is only meaningful under stated constraints.
- Markers are made manually (traced on paper or fabric) or by CAD, and duplicated by carbon, perforation or copying; waste inside the marker is distinct from spreading and cutting losses.

9. Practice

Answer in your own words; brief model answers follow.

1. A marker is 7.2 m long on fabric 1.5 m wide. The pattern pieces total 9.18 m². Calculate the marker efficiency and the waste percentage.
2. A marker of length 5.8 m contains the pieces of 5 complete garments. What fabric length is consumed per garment, and what other information would you need to express consumption as an area?

3. A buyer demands at least 85% efficiency on a marker whose area is 12 m². What is the maximum permitted waste area, and the minimum pattern area?
4. A velvet order and a plain cotton order use the same patterns. Explain why the velvet marker will show lower efficiency, naming the constraint responsible.
5. A CAD marker reports 91% efficiency, but the style is a bold check. Why should the technologist question the figure before accepting it?
6. Distinguish marker waste from spreading end loss, and state why a factory reports them separately.

Answer key

1. Marker area = 7.2 × 1.5 = 10.8 m². Efficiency = 9.18 ÷ 10.8 × 100 = 85.0%. Waste = 15.0%.
2. 5.8 ÷ 5 = 1.16 m per garment. For area, the usable fabric width is needed: consumption area = 1.16 × width.
3. Minimum pattern area = 0.85 × 12 = 10.2 m²; maximum waste = 12 − 10.2 = 1.8 m².
4. Velvet has a pile (nap) direction: every piece must point the same way, so pieces cannot be rotated 180° to interlock; the nap-direction constraint removes the freedom that produces tight nesting.
5. Because check matching forces pieces to positions set by the fabric repeat; a 91% plan very likely ignored matching, so cutting it would produce unmatched checks — the figure is only valid under the matching constraint.
6. Marker waste is the uncovered area inside the marker plan (inter-pattern, end and width loss), a property of the plan. Spreading end loss is fabric lost at the ends of plies while building the lay, a property of execution. Separate reporting shows whether to fix the planning or the spreading.

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